CIRCUIT DEBUGGING ASSISTANT

Electronic Circuit Fault Localization & Diagnostic Report

Report ID: ANALYSIS-C1895939 **Date:** 2026-10-13 22:55:36

Circuit Type: Rc Low Pass **Diagnostic Engine:** Mahajan et al. Hybrid Core v1.0

1. Executive Summary

This report documents the automated topological analysis, numerical simulation, and evidence-grounded fault localization conducted on the uploaded schematic. The device under test (DUT) was classified as a **Rc Low Pass**. The deterministic multi-evidence residual engine identified a primary anomaly on **C +20%** with a normalized residual score of **0.0000**. A total of 0 topological or extraction findings were cataloged.

2. Scope and Objective

The objective of this pipeline is to provide an auditable, physically grounded fault localization framework. Unlike black-box classifiers, the numerical core computes normalized residuals over multi-domain observables (frequency transfer response and independent current/transient observables) to eliminate parameter ambiguity.

3. Extracted Circuit Components and Netlist

Ref Designator	Component Type	Nominal Value	Tolerance	Terminals	Confidence
R	Resistor	1000 Ω	±5.0%	out, 0	0.98
C	Capacitor	120.00 nF	±10.0%	out, 0	0.97
Vin	DC/AC Source	1 V	±1.0%	in, 0	1.00 (Verified)

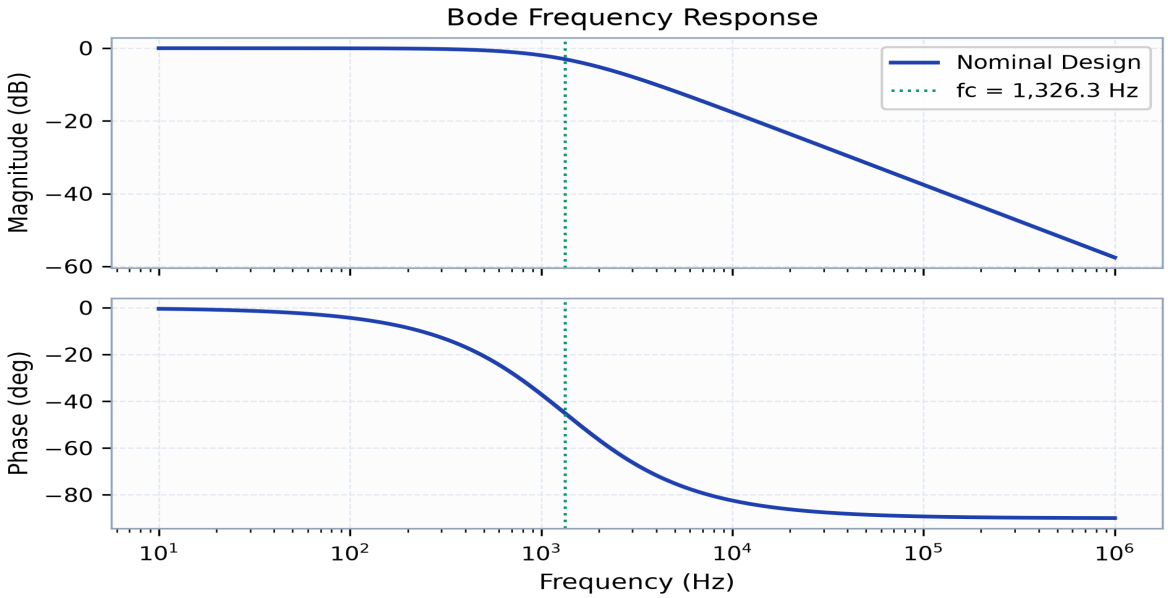
4. Topological and Consistency Findings Register

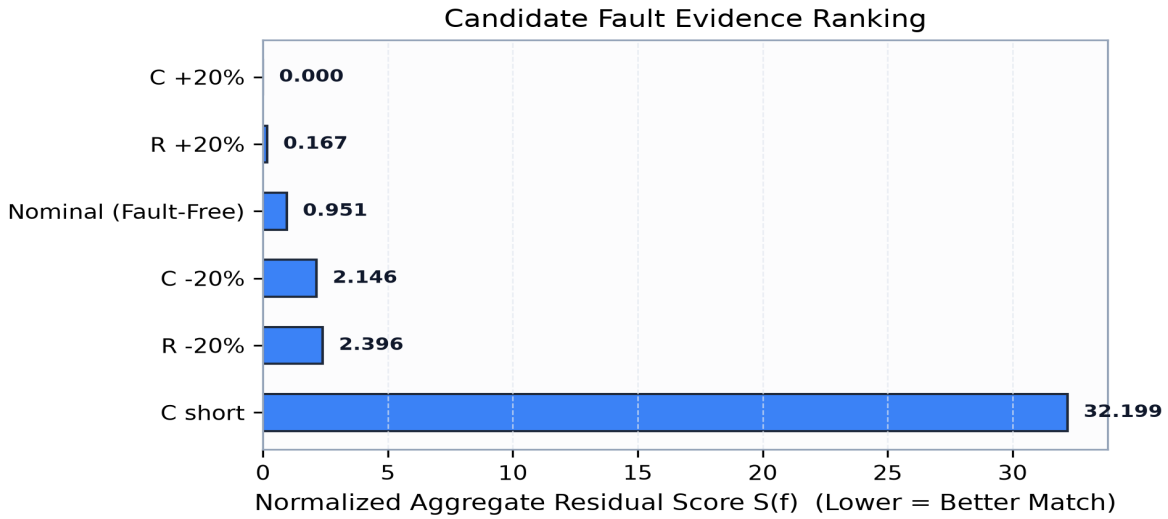
No topological or netlist consistency violations detected.

5. Governing Circuit Equations

• Complex Transfer Function: $H(f) = 1 / [1 + j 2\pi f R C]$
• Cutoff Frequency: $f_c = 1 / [2\pi R C]$
• Independent Initial Step Current: $I_{step}(0+) = Vin / R$ (Sensitive to R, independent of C)

6. Simulation Curves & Residual Diagnostic Evidence





7. Adaptive Test-Point Recommendation (Value-of-Information U(t))

Recommended Action: Perform **DMM Resistance Measurement on R1**.

• Relative Measurement Cost C(t): 1.5

• Expected Entropy Reduction ΔH: 0.320 bits

• Utility Metric U(t): **0.213**

Technical Rationale: This physical measurement maximally separates ambiguous fault hypotheses with minimal bench exploration effort.

8. Reproducibility Appendix & Limitations

Limitations: Evaluated on analytical linear circuit models and deterministic equations. Real physical breadboards or PCBs introduce parasitic inductance, contact resistance, and component tolerances.

Reproducibility Information:

• Framework Version: Mahajan et al. Circuit Debugging Assistant v1.0.0

• Python Environment: 3.13 | NumPy | SciPy | Matplotlib | ReportLab

• SPICE Compatibility: Standard ngspice / Berkeley SPICE 3f5 netlist format exported.

• Reference Paper: S. Mahajan, S. Khalkar, S. Pathan, 'Circuit Debugging Assistant: A Hybrid Simulation- and Evidence-Driven Framework for Electronic Circuit Fault Localization', KKWIEER Nashik, 2026.